

The Red/Black Game:

Rules, Deterministic Dynamics, and Exact State Enumeration

(Heads-Up)

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Abstract

Red/Black is a two-player bidding game played over a shared 52-card deck in which only color matters. We give a complete formal account of the heads-up game and count everything in it exactly. The account is organized as a ladder of four descriptions of where the game stands, in decreasing order of detail. First, the *dynamics state*: hands plus the current round’s bids and flips. This is the state that drives the rules; it is finite, deterministic once the deal is fixed, and acyclic, so every game lasts five to nine rounds. Second, the *history*: the full public record since the deal, including every past round’s bidding. Strategies and beliefs may condition on all of it, so it is the true state of the game G , and it is enormous: 5.3×10^{34} public decision histories and exactly 2,448,351,139,319,171,077,272,326,424,233,304 perfect-recall information sets at hand size five. Third, the *public class*: two certainty bounds per seat plus the identity of the pending discarder, 6,051 classes of which 5,485 are reachable. The class records exactly what is publicly known (it pins the support of any posterior) and none of what is believed (its shape). Fourth, the *bridge* between the second and third rungs: under any strategy profile, the posterior at a round boundary factorizes as the class prior times two per-seat likelihood tilts, so the belief-relevant content of an arbitrarily long history compresses without loss into the class and two short vectors. The companion papers stand on the two ends of this ladder: Paper II solves the abstraction G' obtained by forgetting the tilts and measures what the forgetting costs; Paper III carries the tilts and pursues a certified bound for G itself. Every count is machine-verified against the shipped rules engine.

1 Introduction

Red/Black is a bidding game for two to ten players. A shared 52-card deck is dealt face down; only the color of a card matters, and there are exactly 26 red and 26 black. Players bid on how many cards of a chosen color are in play across *all* hands. An opponent either raises the bid or challenges it, and a challenged bidder must literally produce the claimed cards by flipping them, one at a time, off the top of any player’s deck. Fail, and you discard a card. Run out of cards and you are eliminated.

The game is small enough to be understood completely and large enough to be interesting. This paper describes the game exactly, formalizes its dynamics as a finite transition system, and counts everything. Strategy, equilibrium and the algorithms that approximate it are the subject of two companion papers; this paper builds the objects those algorithms run on.

The account is organized as a ladder of four descriptions of the state of a game in progress, in decreasing order of detail: the *dynamics state* that drives the rules (hands plus the current round’s bids and flips); the *history*, on which strategies and beliefs may lawfully condition, and which is the true state of the game G ; the *public class*, a small integer summary that captures exactly what is publicly known; and the *rank-one bridge*, by which the belief content of a history compresses losslessly to the class plus two per-seat tilt vectors. Section 3 states the ladder precisely and points each rung to the section that develops it.

Scope. We restrict throughout to the **heads-up** ($N = 2$) game. The rules in Section 2 are stated for two players; where a rule generalizes awkwardly we say so. All counting is heads-up.

Contributions.

1. A complete formal specification of Red/Black as a deterministic, acyclic transition system (Section 4).
2. Termination bounds: the heads-up game lasts between 5 and 9 rounds and never revisits a state (Theorems 4.7 and 4.8).
3. An order-irrelevance lemma (Lemma 5.3) collapsing procurement from a tree of flip sequences to a lattice on two counters (Section 5).
4. A closed form $\binom{H+3}{3} - 1$ for the per-seat public statistic, hence 6,051 public classes at hand size $H = 5$ (Section 6).
5. An exact reachability closure: 5,485 of the 6,051 classes occur, with the obstruction identified and proved (Section 8.2).
6. Support completeness: the class pins the support of the public posterior exactly, though not its shape (Theorem 10.1).
7. Exact size of the true game G : 5.3×10^{34} public decision histories and 2.4×10^{33} perfect-recall infosets at $H = 5$, counted rather than estimated (Section 9).
8. The rank-one factorization of reachable beliefs (Theorem 10.3), with opponent-model-free own-side conditionals (Corollary 10.4).

Reproducibility. Two scripts, `state_counts.py` and `real_game_size.py`, regenerate every numeral in this paper from the rules, independently of the game engine. `verify_against_engine.py` cross-checks the counts against the shipped engine. All three scripts accompany this document (Appendix A).

Notation. Symbols are fixed for the whole paper and collected in Table 1. Seats are always 0 and 1, and i always ranges over them. Two conventions prevent collisions: lower bounds always carry the lb subscript, so b_i (blacks held) is never confused with b_{lb}^i (blacks known to be held); and bid counts are written k or κ , never b . A primed symbol always means “after the pending discard”; the distinction is the subject of Section 6.2, and it is the one place where the rules’ ordering and the analysis’ ordering disagree.

2 The Rules

2.1 Components and setup

One shared deck of 52 cards, exactly 26 red and 26 black. Suits and ranks do not exist: a card’s only attribute is its color. Each of the two players is dealt $H = 5$ cards; the remaining $52 - 2H = 42$ cards are removed from the game and never seen by anyone. Hands only ever shrink; no card is drawn after the deal.

Table 1: Notation used throughout.

Symbol	Meaning	Defined
N	number of players; $N = 2$ throughout	§2
H	hand size at the deal, and the most a seat can ever hold	§2
n_i	cards in seat i 's hand <i>now</i> (at a boundary: pre-discard)	Def. 4.1
T	cards in play now, $T = n_0 + n_1$	Def. 4.1
n'_i	seat i 's hand size <i>during</i> the next round (post-discard)	§6.2
T'	cards in play during the next round; the layer index	§6.2
r_i, b_i	reds and blacks currently in seat i 's hand, $r_i + b_i = n_i$	§8.3
ρ	phase	Def. 4.1
σ	seat that starts the current round	Def. 4.1
h_i	seat i 's hand as an <i>ordered</i> tuple of colors	Def. 4.1
β	bid history: a color plus a chain $k_1 < \dots < k_m$	Def. 4.1
ν, λ	the bidder; the loser of the contest just resolved	Def. 4.1
φ_i	cards flipped off seat i 's deck this contest	Def. 4.1
c	the bid color of the current round	§2
k, κ	a bid count; the <i>final</i> count of a chain ($\kappa = k_m$)	§2
x, y	flips taken off the bidder's / the challenger's deck	Lem. 5.6
L	lattice-point count for procurement	Lem. 5.6
ℓ	seat owing a discard at a boundary; -1 at the root	Def. 6.1
s_i	seat i 's public statistic ($n_i, r_{\text{lb}}^i, b_{\text{lb}}^i$)	Def. 6.1
$r_{\text{lb}}^i, b_{\text{lb}}^i$	certainty bounds: reds / blacks seat i surely holds	Def. 6.1
$K(H)$	number of distinct per-seat public statistics	Thm. 6.6
m_i, e_i	on-color / off-color cards revealed from seat i in a round	Lem. 8.2

A player's hand is an *ordered* list. Index 0 is the top of the deck: the first card that will be flipped if someone procures from it. Players see their own cards and nothing of their opponent's except its size, which is always public.

2.2 A round

A round has four phases, in order.

1. **Reorder.** Both players privately and simultaneously rearrange their own hand and confirm. This is the only moment hand order can change.
2. **Bidding.** The starting player bids a pair (k, c) : a count $k \geq 1$ and a color c , claiming that at least k cards of color c are in play across both hands. The count is capped by the total number of cards in play. Thereafter players alternate, and each must do exactly one of: *raise* (same color, strictly greater count) or *challenge*. The color is fixed by the first bid and never changes within a round. Passing is not a legal move.
3. **Procurement.** A challenge ends bidding. The last bidder must now produce the claim. They repeatedly flip the top card of *either* player's deck (their own or the opponent's, switching freely), and each flipped color is revealed publicly. The first off-color flip is fatal: the bidder loses immediately. Reaching the claimed count of on-color cards wins: the challenger loses.
4. **Discard.** The loser of the contest secretly removes one card from their hand. The identity

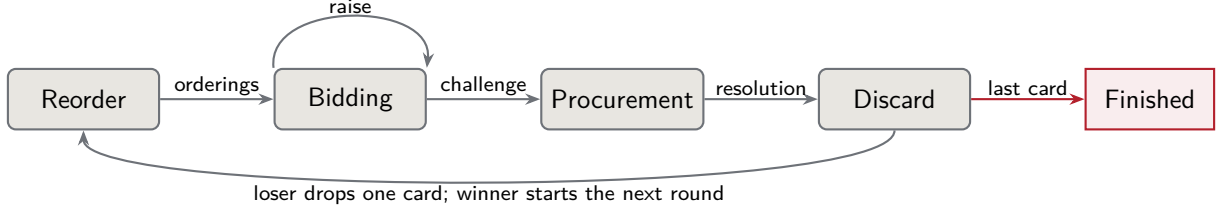


Figure 1: The four phases of a round. The only cycle in the *phase* graph is the round loop; it is not a cycle in the state graph, because one card leaves play on every pass (Theorem 4.7).

of that card is never revealed. If the loser holds exactly one card, the discard is forced and eliminates them.

The winner of the contest starts the next round. In round 1 the starting player is seat 0. A player with zero cards is eliminated and the other player wins.

2.3 Information

What is public at all times: both hand sizes, whose turn it is, the current round’s full bid history, and the full flip history of the current contest. What is private: the colors and order of your own hand, and the identity of any card you yourself discarded.

Two rules do most of the work in this paper.

Discard anonymity. The discarded card’s identity is never revealed. Observers learn only that a hand shrank by one.

Reorder scrambling. Cards flipped in round k are identifiable only within round k . Once the next reorder happens, all of a player’s cards are anonymous card backs again.

Together these mean no *card identity* survives a round boundary. What survives is strictly weaker: multiset knowledge, degraded by discards. If you saw two reds flipped from my hand, you know I hold at least two reds until I discard; after that you know only that I hold at least one, because the card I dropped may have been one of them. We make this precise in Section 6.

3 Four Descriptions of the State

The paper is organized around four descriptions of where a game of Red/Black stands. They are listed here in decreasing order of detail, and each later section develops one rung.

(i) The dynamics state. The tuple that makes the rules run: both hands, the current round’s bid chain, and the flip record of the current contest. This is the minimal state for which every transition is a function of the state and the acting player’s move. Past rounds genuinely do not affect the dynamics: a cleared bid chain and a departed discard change no legal move and no transition. Section 4 formalizes this state, proves the transition system deterministic and acyclic, and derives the round bounds; Sections 5 and 6 count its round structure and its public projection.

(ii) **The true game state.** The full public history since the deal, including every past round’s bidding and flips, paired with each seat’s private trajectory. Nothing in the rules erases that record from the players’ minds, so strategies and beliefs may lawfully condition on all of it. This is the state notion of the real game G , and its count is the headline of the paper: 5.3×10^{34} public decision histories and 2.448×10^{33} perfect-recall information sets at hand size five (Section 9).

(iii) **The public class.** Two certainty bounds per seat plus the identity of the pending discarder. The class is support-complete: it determines exactly which hands are possible (Theorem 10.1). It is belief-incomplete: it does not determine how likely each possible hand is. It is the state notion of the abstraction G' (Definition 10.6) that Paper II solves. Sections 6 and 8.2 define and count it: 6,051 classes, of which 5,485 are reachable.

(iv) **The bridge.** A rank-one factorization connects rungs (ii) and (iii): under any strategy profile, the posterior at a round boundary is the class prior times two per-seat likelihood tilts (Theorem 10.3). The belief-relevant content of an arbitrarily long history therefore compresses losslessly to the class plus two short vectors. Section 10 proves this and measures what discarding the tilts costs.

Paper II solves G' , and Paper III carries the tilts toward a certified bound for G itself.

4 The Dynamics as a Deterministic Transition System

This section builds rung (i): the state that drives the rules.

4.1 Determinism and hidden information

A deterministic finite automaton is a tuple $(\mathcal{Q}, \Sigma, \delta, q_0, F)$ with $\delta : \mathcal{Q} \times \Sigma \rightarrow \mathcal{Q}$ a *function*: one input, one successor, no coin flips. The phrase “hidden information” suggests Red/Black cannot be such a machine, but it is, once two points are settled. First, the randomness sits in the initial state rather than the transitions. The deal is the only chance event in the entire game; fix it, and every subsequent transition is a deterministic function of the current state and the acting player’s move, so Red/Black is a family of deterministic automata indexed by the deal. Second, hidden information is a property of the observers rather than of the machine. The state is well defined at all times even though neither player can see all of it; formally, hidden information is an equivalence relation on $\mathcal{Q}$ (Section 4.6), layered on top of a machine that is itself deterministic.

4.2 The state

Definition 4.1 (Dynamics state (Markov state)). A state is a tuple

$$q = (\rho, \sigma, h_0, h_1, \beta, \nu, \varphi_0, \varphi_1, \lambda)$$

with components

ρ	phase, $\rho \in \{\text{REORDER, BID, PROC, DISC, FIN}\}$
σ	starting seat of the current round, $\sigma \in \{0, 1\}$
h_i	seat i 's hand: an ordered tuple of colors, $h_i \in \{\text{R, B}\}^{n_i}$
β	bid history: $\perp$, or a pair $(c, k_1 < \dots < k_m)$ of a color and a chain of counts
ν	the bidder (last raiser) once a challenge has occurred, else $\perp$
φ_i	cards already flipped from seat i 's deck this contest
λ	the loser of the contest just resolved, else $\perp$

We write $n_i = |h_i|$ for hand sizes and $T = n_0 + n_1$ for the total cards in play.

Remark 4.2 (What the dynamics state omits, and why). The state carries no record of *which* cards have been discarded, nor of past rounds' bids or flips, because no transition reads them: discarded cards are gone from play, and a bid chain is cleared when its round resolves. Omitting them is what makes this state space finite and small. They are not absent from the *game*. Players remember past bids and flips and may condition strategy and belief on them, so they belong to the true state of G , rung (ii) of Section 3; Section 9 counts what keeping them costs.

4.3 The input alphabet

Definition 4.3 (Inputs). $\Sigma = \Sigma_{\text{deal}} \cup \Sigma_{\text{ord}} \cup \Sigma_{\text{bid}} \cup \Sigma_{\text{proc}} \cup \Sigma_{\text{disc}}$, where

Phase	Letters available	How many
—	deal (D), a split of $2H$ cards	$\binom{52}{H} \binom{52-H}{H}$
REORDER	order (π), an ordering of one's own hand	$\binom{n_i}{r_i}$ per seat
BID	bid (c, k), $c \in \{\text{R, B}\}$, $1 \leq k \leq T$	$2T$
BID	raise (k), k above the current count	$\leq T - 1$
BID	challenge	1
PROC	flip (j), j the target seat	≤ 2
DISC	drop (c), $c \in \{\text{R, B}\}$	≤ 2

Opening bids and raises are mutually exclusive: **bid** is legal only on an empty history, **raise** and **challenge** only on a non-empty one.

δ is a *partial* function: illegal letters have no successor. One may complete it in the usual way with an absorbing dead state; we leave it partial, since the legal-move set is itself part of the game's description.

4.4 Initial and final states

Definition 4.4 (Initial states). q_0 is the pre-deal state. After the single chance letter **deal**(D) the machine occupies $(\text{REORDER}, 0, h_0, h_1, \perp, \perp, 0, 0, \perp)$ with $|h_0| = |h_1| = H$. Seat 0 starts round 1 by convention.

Definition 4.5 (Final states). F is the set of states with $\rho = \text{FIN}$, equivalently those in which some $n_i = 0$. The other player is the winner. There are no draws: exactly one player is eliminated, and the game ends the moment that happens.

4.5 Determinism, acyclicity, termination

Theorem 4.6 (Determinism). *For every state q and every legal letter a , the successor $\delta(q, a)$ is uniquely determined. No transition consults a random source.*

Proof. Inspect each phase. REORDER replaces h_i by a stated permutation. BID appends to β or moves to PROC and records ν . In PROC, the letter names a target seat j ; the card revealed is $h_j[\varphi_j]$, already fixed in the state, and the outcome is a comparison of that color against β 's color: off-color $\Rightarrow$ bidder loses, else increment φ_j and check the count. DISC names a color and removes one such card. In every case the successor is a function of (q, a) alone. $\square$

The subtle point is in PROC: the flipped card looks random to the players, but it was fixed at the deal and positioned at the last reorder. The uncertainty is epistemic rather than stochastic.

Theorem 4.7 (Acyclicity). *The transition graph is acyclic.*

Proof. Take the lexicographic rank $\Phi(q) = (T, -|\beta|, -(\varphi_0 + \varphi_1))$ ordered so that every transition strictly decreases it. Within a contest, each **flip** increments $\varphi_0 + \varphi_1$ and leaves T and $|\beta|$ fixed. Within bidding, each **bid** or **raise** increments $|\beta|$ and leaves T fixed; **challenge** moves to PROC with $\varphi_0 + \varphi_1 = 0$, which is handled by treating phase order as a tiebreak. Each **drop** decrements T by exactly one, resetting the later components. As T strictly decreases once per round and never increases, no state recurs. $\square$

Theorem 4.8 (Round bounds). *The heads-up game with hand size H lasts at least H and at most $2H - 1$ rounds. For the real game ($H = 5$) that is between 5 and 9 rounds.*

Proof. Exactly one card leaves play per round, and a seat is eliminated on its H -th discard. The minimum is achieved when the same seat loses every round: H rounds. For the maximum, the game is still alive after k rounds only if both seats have made at most $H - 1$ discards, so $k \leq 2(H - 1)$; the game therefore ends on round $2H - 1$ at the latest, and that bound is achieved by alternating losses. $\square$

Corollary 4.9. *Total cards in play is a strictly decreasing round counter running $2H, 2H - 1, \dots$ down to the terminal value, so it is a natural index for induction over the game. Section 6.2 fixes precisely which version of that count we index by, the one measured during a round rather than at its boundary, and calls it T' .*

4.6 The information partition

Definition 4.10 (Observation and indistinguishability). Let $\text{obs}_i(q)$ be the part of q visible to seat i : the phase, the starter, both hand sizes, the current bid history β , the flip counters and the colors revealed so far this contest, plus seat i 's own hand h_i . Write $q \sim_i q'$ when $\text{obs}_i(q) = \text{obs}_i(q')$. The classes of $\sim_i$ are seat i 's *information sets*.

$\sim_i$ is what makes Red/Black a game rather than a puzzle, and it is invisible to $(\mathcal{Q}, \Sigma, \delta, q_0, F)$. Perfect recall holds: a player never forgets their own past observations or moves. The public projection $\text{pub}(q)$, the part of q both players see, is far smaller than q , collapsing at round boundaries to just four integers (Section 6). The information sets of the full game G are equivalence classes of *histories* rather than of dynamics states; Section 9 counts them.

5 The Round Graph

Before counting whole games we isolate the structure of a *single* round, which is independent of the deal and depends only on the starter σ and the hand sizes (n_0, n_1) .

5.1 Bidding

A bid history is a color together with a strictly increasing chain of counts drawn from $\{1, \dots, T\}$. Chains are in bijection with non-empty subsets of $\{1, \dots, T\}$.

Lemma 5.1 (Bid node count). *The number of bidding states with T cards in play is*

$$1 + 2(2^T - 1) = 2^{T+1} - 1.$$

Proof. One pre-bid state; then a state per (color, non-empty chain) pair, and there are $2^T - 1$ non-empty subsets of $\{1, \dots, T\}$ and two colors. $\square$

So $T = 2$ gives 7 states, $T = 4$ gives 31, and the full game's opening round ($T = 10$) gives 2,047. The seat to act at a chain of length m is σ if m is even and $1 - \sigma$ otherwise; the bidder, the seat that will have to produce the claim, is whoever placed the last bid.

Remark 5.2 (A constraint used later). Because the starter bids first and counts strictly increase, a *non-starter* can only be the bidder if at least two bids were placed, which forces the final count to be at least 2. A final count of 1 pins the bidder to be the starter. This small fact prunes real states in Section 8.2.

5.2 Procurement, and why order does not matter

Lemma 5.3 (Order irrelevance). *During a contest, the state depends only on the pair of counters (φ_0, φ_1) and not on the order in which the flips were made.*

Proof. A contest continues only while every flipped card has matched the bid color. Hence at any live procurement state, the multiset of revealed cards is determined: $\varphi_0 + \varphi_1$ cards, all of the bid color, of which φ_i came off seat i 's deck. Each deck is consumed strictly top down, so which cards remain is likewise determined by the counters alone. Two flip sequences with the same counters therefore yield identical states. $\square$

Lemma 5.3 reduces procurement from a tree of flip sequences to a *lattice* on two counters. A naive sequence count at $T = 10$ would be exponential in the number of flips; the lattice is quadratic.

A second, independent collapse applies to the *orderings* themselves.

Lemma 5.4 (Hand orderings collapse at the first color change). *Let a seat's post-reorder hand open with a maximal run of color a of length k . Then nothing at position $k+2$ or beyond can ever be revealed, under either bid color. Consequently two orderings are strategically indistinguishable whenever they agree on the pair*

$$(a, k) = (\text{leading color}, \text{leading run length}),$$

and the number of distinguishable orderings of an (r, b) multiset is

$$F(r, b) = \begin{cases} 1, & r = 0 \text{ or } b = 0, \\ r + b, & \text{otherwise,} \end{cases}$$

against $\binom{r+b}{r}$ raw orderings.

Proof. Procurement from a seat's deck consumes cards top down and stops at that seat's first off-color card. If the bid color is a , the flips take the k cards of the leading run and then the card at $k + 1$, which is $\neg a$ by maximality, and which is fatal. If the bid color is $\neg a$, the very first flip is off-color and fatal. In both cases consumption halts at position $k + 1$ at the latest, so positions $k + 2, \dots, n$ are never seen.

Nor can they matter later. Flipping does not remove a card (Section 2), so a round leaves hand *composition* untouched; and every round begins with a fresh reorder, so no ordering survives a boundary. Hence the tail is unobservable and irrelevant, and the observable content of an ordering is exactly (a, k) .

For the count: with both colors present, the leading run may be $a = \text{red}$ of length $1..r$ or $a = \text{black}$ of length $1..b$, giving $r + b$ classes; a single-color hand admits one. $\square$

Remark 5.5 (Size of the collapse). The saving needs $n \geq 4$ to appear at all (for $n \leq 3$, $\binom{n}{r} = F$ exactly) and is capped at 50% per composition ($\binom{5}{2} = 10 \rightarrow 5$); lopsided hands ($r \in \{0, 1, n - 1, n\}$) save nothing. Summed over every reachable private state of the full game the reduction is 12% ($172,488 \rightarrow 150,964$), growing with depth from 0% at $T' \leq 4$ through 15% at $T' = 8$ and 24% at $T' = 9$ to 31% at the root; weighted by how many decisions sit in each layer it is about 19% (Paper II tabulates the weights). The collapse is *lossless*, like Lemma 5.3 and unlike the belief abstraction of Section 10: merged orderings are indistinguishable in every continuation.

Lemma 5.6 (Procurement node count). *Fix a color and a chain $k_1 < \dots < k_m$, and write n_{bid} and n_{chal} for the hand sizes of the bidder and the challenger. The live procurement states for that chain number*

$$L(k_m - 1, n_{\text{bid}}, n_{\text{chal}}) = \#\{(x, y) \in \mathbb{Z}_{\geq 0}^2 : x + y \leq k_m - 1, x \leq n_{\text{bid}}, y \leq n_{\text{chal}}\},$$

where x and y count the cards already flipped off the bidder's and the challenger's decks respectively. Summing over both colors and all 2^{k_m-1} chains with final count k_m gives the round's total. When $n_0 = n_1 = n$ this is

$$\#PROC(T, n) = 2 \sum_{\kappa=1}^T 2^{\kappa-1} L(\kappa - 1, n, n),$$

the sum running over the possible final counts κ .

The constraint $x + y \leq k_m - 1$ is the statement that the bid is not yet made: at $x + y = k_m$ the bidder has produced the claim and the round is over.

5.3 Resolutions

Every contest ends in one of two ways, and the resolution records enough to continue the game: the color and chain, the final counters, whether the loser was the bidder, and, when the bidder busted, which deck the fatal card came off. That last datum matters because it tells observers whose hand certainly contains an off-color card.

Table 2 collects the three counts for the hand-size configurations used later.

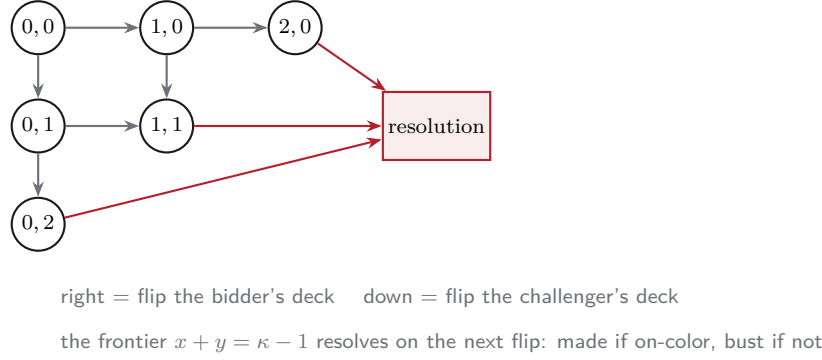


Figure 2: The procurement lattice for a final bid count of $\kappa = 3$ with two cards in each hand. A state is the pair (x, y) counting cards already flipped off the bidder's and the challenger's decks; by Lemma 5.3 the path taken to a node is irrelevant. Only the $x + y \leq \kappa - 1 = 2$ nodes are live.

Table 2: The round graph, per (n_0, n_1) . Counts are independent of the starter, and every entry is verified against the shipped graph builder.

Configuration		T	Bid states	Procurement states	Resolutions
(1, 1)	one-card game	2	7	14	28
(2, 2)	two-card game	4	31	190	348
(5, 5)	opening round	10	2,047	65,790	119,036
(3, 5)	a later round	8	511	10,398	18,588

6 The Public Class at a Round Boundary

We now derive the projection $\text{pub}(q)$ that both players share.

6.1 What survives a round

Fix the moment a contest resolves: the loser is known, but has not yet discarded. Call this a *boundary*. By the rules of Section 2.3, an observer at a boundary knows both hand sizes, who lost, and, from the flips just seen plus memory of earlier rounds, a lower bound on how many cards of each color each seat certainly still holds. Nothing else about cards survives, because discard anonymity and reorder scrambling destroy identity.

Definition 6.1 (Public class). A boundary's public state is

$$(\ell, s_0, s_1), \quad s_i = (n_i, r_{\text{lb}}^i, b_{\text{lb}}^i),$$

where $\ell \in \{-1, 0, 1\}$ names the seat that owes a discard (-1 at the game root), n_i is seat i 's hand size *before* that discard, and $r_{\text{lb}}^i, b_{\text{lb}}^i$ are the certainty bounds: everyone knows seat i holds at least r_{lb}^i reds and at least b_{lb}^i blacks. The next round's starter is $1 - \ell$, so it need not be stored.

Lemma 6.2 (Bound folding). *Across a round, the bounds evolve as*

$$r_{\text{lb}} \mapsto \max(r_{\text{lb}} - [\text{this seat discarded}], \#\{\text{reds seen from this seat this round}\}),$$

and symmetrically for b_{lb} . The fold is a maximum, never a sum.

Proof. A discard removes an unknown card, so a bound of r_{lb} degrades to $r_{\text{lb}} - 1$ (floored at zero): the dropped card may have been one of the counted reds. Cards flipped this round are certainly still held, giving a fresh bound. The two cannot be added, because a red seen this round may be one of the reds already counted; only the larger is certain. $\square$

Lemma 6.3 (Certainty is bounded by holdings). $r_{\text{lb}}^i + b_{\text{lb}}^i \leq n_i$ at every boundary.

6.2 Two clocks: where the discard sits

There is one genuine offset in this paper, and it causes trouble if left implicit, so we make it explicit here and use it consistently afterwards.

The rules order a round as reorder $\rightarrow$ bidding $\rightarrow$ procurement $\rightarrow$ *discard*: the discard is the *last* thing that happens (Section 2, Figure 1). That is the right description of the game as played.

But the natural place to cut the game is not the same place. We cut at a *boundary*, a procurement resolution where the loser is known but has not yet discarded, because that is the moment at which the public state is a clean summary. Cutting there means the segment between two consecutive boundaries reads *discard* $\rightarrow$ reorder $\rightarrow$ bidding $\rightarrow$ procurement: the discard is now the *first* thing that happens.

Nothing about the game changed. The same events occur in the same order; only the cut points moved, by exactly one discard.

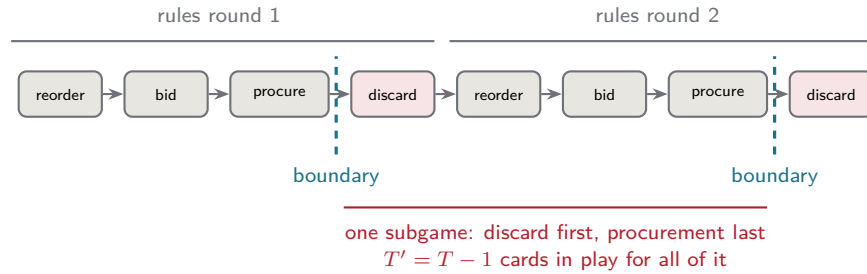


Figure 3: The same event tape, cut two ways. The rules bracket a round from reorder to discard (grey, above). The analysis brackets a subgame from one procurement resolution to the next (red, below), so the discard moves to the front. A boundary sits between procurement and discard, which is why a boundary's hand sizes n_i are *pre*-discard while the round that follows is played with n'_i .

Definition 6.4 (In-round sizes and the layer index). At a boundary with pending discarder ℓ , write

$$n'_i = n_i - [i = \ell], \quad T' = n'_0 + n'_1 = T - [\ell \geq 0].$$

So n_i and T describe the table at the boundary; n'_i and T' describe it for the entire round that follows, since no card leaves play between the discard and the next resolution. We index classes by T' , the **layer**.

Remark 6.5 (Why the layer is T' and not T). Two boundaries with the same T can have different amounts of play left: the game root has $T = 10$ with no discard pending, whereas a resolution at $n_0 = n_1 = 5$ also has $T = 10$ but owes a discard, so only 9 cards will actually be bid over. Indexing by T' puts them in different layers, which is what makes the class space a clean DAG with exactly one edge per round (Figure 6). Indexing by T would not.

Two consequences are easy to misread, so we state them plainly.

- A boundary's hand sizes n_i are *pre-discard*. A seat listed with $n_i = 1$ is about to be eliminated, not already gone.
- Everything in Section 5, the round graph, is expressed in the sizes that hold *during* the round. A round reached from a boundary therefore uses n'_i and T' there, not n_i and T . Read in isolation, Section 5 describes a generic round and the primes are unnecessary; read as the successor of a boundary, they matter.

6.3 Counting public classes

Theorem 6.6 (Per-seat statistic). *For hand size H , the number of distinct values of $(n, r_{\text{lb}}, b_{\text{lb}})$ with $1 \leq n \leq H$ and $r_{\text{lb}} + b_{\text{lb}} \leq n$ is*

$$K(H) = \sum_{n=1}^H \binom{n+2}{2} = \binom{H+3}{3} - 1.$$

Proof. For fixed n , the pairs $(r_{\text{lb}}, b_{\text{lb}})$ with $r_{\text{lb}} + b_{\text{lb}} \leq n$ number $\binom{n+2}{2}$ (weak compositions of at most n into two parts). The hockey-stick identity gives $\sum_{n=0}^H \binom{n+2}{2} = \binom{H+3}{3}$; dropping the $n = 0$ term, worth 1, gives the claim. $\square$

So $K(1) = 3$, $K(2) = 9$, $K(5) = 55$.

Corollary 6.7 (Class count). *The number of public classes is $2K(H)^2 + 1$: two choices of the seat owing a discard, an independent statistic per seat, plus the game root. For $H = 5$ that is $2 \cdot 55^2 + 1 = 6,051$.*

Corollary 6.8 (Terminal classes). *A boundary whose pending loser holds one card is terminal: the discard is forced and eliminates them. There are $2 \cdot 3 \cdot K(H)$ such classes, or 330 at $H = 5$.*

Classes are indexed by the layer T' of Definition 6.4. By Theorem 4.7, T' strictly decreases from one boundary to the next, so the class space is a layered directed acyclic graph.

7 Small Games

Two scaled-down games, identical in rules but dealt $H = 1$ and $H = 2$ cards, are small enough to enumerate by brute force. They serve as checkpoints for the closed forms of Sections 5 and 6, for the closure of Section 8.2, and for the inforeset recurrence of Section 9, which is validated against explicit joint enumeration at exactly these two sizes.

7.1 The one-card game

Deal each player a single card from the real 52-card deck. The game lasts exactly one round (Theorem 4.8 with $H = 1$). There are $52 \cdot 51 = 2,652$ card-level deals and four color outcomes, with

$$\Pr[\text{both red}] = \Pr[\text{both black}] = \frac{26 \cdot 25}{52 \cdot 51} = \frac{25}{102}, \quad \Pr[\text{different}] = \frac{2 \cdot 26 \cdot 26}{52 \cdot 51} = \frac{26}{51} \approx 0.5098.$$

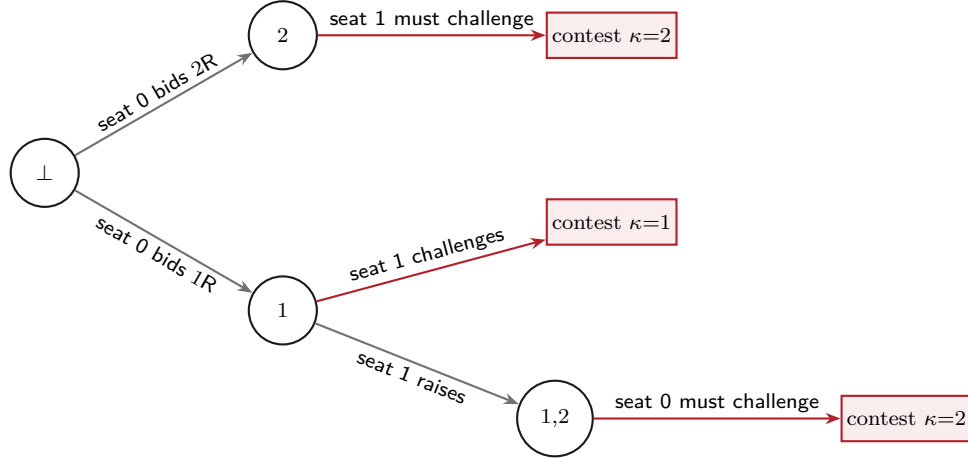


Figure 4: The complete bidding phase of the one-card game, red claims only (black is the mirror). Node labels are the chain of counts; $\perp$ is the pre-bid state. Together with the black mirror this is $1 + 2 \cdot 3 = 7$ states, matching Lemma 5.1.

The two cards are slightly *anti*-correlated, because they are drawn without replacement from a balanced deck; the same effect seeds all card-counting in the full game. With $T = 2$, Lemma 5.1 gives 7 bidding states, and procurement adds 14 states and 28 resolutions (Table 2). Figure 4 draws the whole bidding phase for one color; the other color is an exact mirror. On the public side, Theorem 6.6 gives $K(1) = 3$ and Corollary 6.7 gives 19 classes; exhaustive forward search (Section 8.2) finds 28 concrete boundary states realizing 17 of the 19.

Example 7.1 (The two impossible classes). The two unreachable classes are $s_0 = s_1 = (1, 0, 0)$, for either loser: a boundary at which nothing at all is publicly known about either card. It cannot occur, because a contest cannot resolve without at least one public flip (Lemma 8.1). By contrast $s_0 = s_1 = (1, 0, 1)$, both seats known to hold a black, might look impossible on the grounds that under a red claim only one black can ever be revealed. The claim need not be red: under a black claim of 2, the bidder flips one black off each deck, makes the bid, and both bounds rise together. Fixing the bid color too early is the standard way to get such arguments wrong (Remark 8.5).

7.2 The two-card game

At $H = 2$ three new features appear: a bidding ladder with room to maneuver, a discard that is a genuine choice rather than a forced elimination, and a second round, the first time the game recurses into a smaller copy of itself. There are $\binom{52}{2} \binom{50}{2} = 1,624,350$ card-level deals and 9 color outcomes $(r_0, r_1) \in \{0, 1, 2\}^2$, and the game lasts 2 or 3 rounds (Theorem 4.8). The opening round has $2^5 - 1 = 31$ bidding states, 190 procurement states and 348 resolutions, matching Table 2. On the public side, $K(2) = 9$ gives $2 \cdot 81 + 1 = 163$ classes; forward search realizes 149 of them across 345 concrete boundary states (Table 5).

The round-1 discard is the game's first genuinely private decision with a future: the choice is unobserved, it determines the loser's hand for round 2, and by discard anonymity the opponent never learns it directly. It is also where the public bounds first decay (Lemma 6.2).

Table 3: The color-deal law at $H = 5$: $\Pr[r_0 = u, r_1 = v]$, where r_i is the number of reds dealt to seat i . Rows are u (seat 0), columns v (seat 1). Mass concentrates on balanced hands, and the slight anti-diagonal tilt is the without-replacement effect.

	$v = 0$	1	2	3	4	5
$u = 0$	0.00034	0.00257	0.00713	0.00901	0.00518	0.00109
1	0.00257	0.01783	0.04505	0.05180	0.02713	0.00518
2	0.00713	0.04505	0.10360	0.10854	0.05180	0.00901
3	0.00901	0.05180	0.10854	0.10360	0.04505	0.00713
4	0.00518	0.02713	0.05180	0.04505	0.01783	0.00257
5	0.00109	0.00518	0.00901	0.00713	0.00257	0.00034

Table 4: Public classes at $H = 5$ by layer T' , the cards in play *during* the round that follows, i.e. after the pending discard (Definition 6.4).

T'	With a subgame	Terminal	Total
1	0	18	18
2	36	36	72
3	132	60	192
4	330	90	420
5	686	126	812
6	1,104	0	1,104
7	1,290	0	1,290
8	1,260	0	1,260
9	882	0	882
10	1 (root)	0	1
Total	5,721	330	6,051

8 The Full Game and Reachability

8.1 Deals and classes

The real game deals $H = 5$. There are

$$\binom{52}{5} \binom{47}{5} = 3,986,646,103,440$$

card-level deals and 36 color outcomes $(r_0, r_1) \in \{0, \dots, 5\}^2$, distributed as in Table 3. The opening round alone has 2,047 bidding states, 65,790 procurement states and 119,036 resolutions.

By Corollary 6.7 there are 6,051 public classes. Table 4 breaks them down by layer, separating the 330 terminal classes, where the pending loser holds one card and the value is decided without further play, from the 5,721 that contain a real subgame.

8.2 The reachability obstruction

Corollary 6.7 counts the classes the *definition* admits. Not all of them can occur.

Lemma 8.1 (Every round reveals something). *At least one card is flipped in every contest, so at every non-root boundary at least one of the four bounds is positive.*

Proof. The claim's count is at least 1. The bidder must keep flipping until either the count is reached, which requires at least one on-color card to be revealed, or an off-color card appears, which is itself a flip. Either way at least one card is revealed, and by Lemma 6.2 the corresponding bound is raised to at least 1. $\square$

Lemma 8.2 (Single-round sighting shape). *Within one round, let c be the bid color. Then the cards revealed from seat i 's deck consist of $m_i \geq 0$ cards of color c together with $e_i \in \{0, 1\}$ cards of the other color, and moreover $e_0 + e_1 \leq 1$.*

Proof. Exactly one color is bid per round. Procurement stops the instant an off-color card is flipped, so across the whole round at most one off-color card is ever revealed, and it is the final flip. It comes off exactly one seat's deck, which gives $e_0 + e_1 \leq 1$ and forces every other revealed card to be of color c . $\square$

Corollary 8.3 (One round of history). *At a boundary reached after exactly one round (layer $T' = 2H - 1$, which is $T' = 9$ in the full game) the bounds satisfy, for some common color c :*

1. $\min(r_{\text{lb}}^i, b_{\text{lb}}^i) \leq 1$ for each seat;
2. both seats' bounds are expressed against the same color c ;
3. at most one seat has a non-zero bound on the off color.

Proof. Immediate from Lemma 8.2 and Lemma 6.2: with only one round played there is nothing to fold against, so the bounds are the sightings. $\square$

Example 8.4 (Three cannot-happen states at $T' = 9$). Corollary 8.3 rules out, among others:

Class	Status	Why
$s_0 = (5, 3, 1), s_1 = (5, 0, 0)$	reachable	red bid, 3 reds then a fatal black, all from seat 0
$s_0 = (5, 2, 2), s_1 = (5, 0, 0)$	impossible	needs two off-color reveals (part 1)
$s_0 = (5, 3, 1), s_1 = (5, 2, 1)$	impossible	needs two off-color reveals (part 3)
$s_0 = (5, 3, 0), s_1 = (5, 0, 3)$	impossible	needs two different bid colors (part 2)

Remark 8.5 (A trap worth flagging). Reachability arguments of this kind are easy to get wrong by fixing the bid color too early. Consider $\ell = 0$ with $s_0 = (5, 1, 0), s_1 = (5, 0, 0)$: seat 0 owes the discard *and* is known to hold a red. Reasoning under a red bid, one concludes that no off-color card was shown, so the bid was made, so exactly one red was produced, so the final count was 1; by Remark 5.2 the bidder was then the starter, seat 0, and the loser must therefore be seat 1. That contradiction makes the class look unreachable, and it is not. Under a *black* bid, seat 0 opens, immediately flips its own top card, and it is red: the contest ends at once, seat 0 is the loser, and the revealed red is exactly the certainty bound. This is why we compute the closure by machine rather than argue it by hand.

The obstruction weakens with depth. After two or more rounds, the maximum-fold of Lemma 6.2 can combine sightings under *different* bid colors: a red bid in round 1 revealing (2, 1) followed by a black bid in round 2 revealing (1, 2) folds to (2, 2), which Corollary 8.3 forbids after a single round. So the constraint bites hardest at the top of the game and all but vanishes below.

8.3 Exact closure

To decide which classes occur we search the game forward from the deal. The search cannot run over classes alone. A class records only what is certain, and what a seat can *reveal* in the coming round depends on what it actually holds: a seat whose class says “at least one red” might hold one red or five, and only in the latter case can five reds be flipped off its deck next round, driving r_{lb} to 5. The search must therefore carry the true hand composition alongside the public bounds.

Definition 8.6 (Concrete boundary state). A *concrete boundary state* is

$$(\ell, (r_0, b_0, r_{\text{lb}}^0, b_{\text{lb}}^0), (r_1, b_1, r_{\text{lb}}^1, b_{\text{lb}}^1)),$$

where r_i and b_i are the reds and blacks seat i genuinely holds and the remaining entries are as in Definition 6.1. It always satisfies $r_{\text{lb}}^i \leq r_i$ and $b_{\text{lb}}^i \leq b_i$: you cannot be certain of more than is there.

Each concrete state determines exactly one class (forget r_i and b_i , keep $n_i = r_i + b_i$ and the bounds), so the map

$$\{\text{concrete boundary states}\} \longrightarrow \{\text{classes}\}$$

is a many-to-one projection, and a class is *reachable* precisely when at least one concrete state above it is. This is why the two counts in Table 5 differ so much: 27,990 concrete states at $H = 5$ collapse onto 5,485 classes, roughly five hands per public description.

The search is then a breadth-first closure: start from every color-deal (r_0, r_1) , and from each concrete state generate every state reachable by one legal round, over every discard the loser could make, every bid color, every legal final count, either seat as bidder, and every way the contest could resolve, subject to Lemmas 8.1 and 8.2. It terminates because T strictly decreases (Theorem 4.7) and the state set is finite.

One thing is deliberately *not* carried: how many of a seat’s own discards were red. By Remark 4.2 that changes no legal move and no public bound, so it cannot affect which classes occur. It does affect what players believe, which is the subject of Section 10.

The result reads in two lines. The reachable set is 90.6% of the enumerable one, so the statistics of Definition 6.1 are very nearly a complete invariant. And essentially all of the slack sits in a single layer: $T' = 9$ is the only layer with exactly one round of history, where Corollary 8.3 is strongest, and by $T' = 8$ two rounds of max-folding have already restored almost everything.

9 The True Game: Histories and Information Sets

Everything counted so far is a state of the dynamics or a projection of one. The decision points of the real game G are neither. Players act on what they have seen, and nothing in the rules erases the record from their minds: a bid changes no card and moves no counter, yet it is observed, remembered, and (Section 10) it changes what a rational opponent believes. The state notion of G is therefore the history, rung (ii) of Section 3, and this section counts it exactly.

9.1 Public histories

Definition 9.1 (Public history). The *public history* at a point of play is the full sequence of publicly observable events since the deal: for each round, a reorder marker, the bid chain as bid

Table 5: Exact reachability. “Enumerable” counts the classes Definition 6.1 admits; “reachable” counts those realized by some deal and some legal play. All three games are closed exhaustively.

Game	Boundary states	Reachable classes	Enumerable	Pruned
$H = 1$	28	17	19	10.5%
$H = 2$	345	149	163	8.6%
$H = 5$	27,990	5,485	6,051	9.4%

$H = 5$, layer T'	Reachable	Enumerable	Pruned
1	16	18	11.1%
2	68	72	5.6%
3	186	192	3.1%
4	412	420	1.9%
5	802	812	1.2%
6	1,096	1,104	0.7%
7	1,284	1,290	0.5%
8	1,256	1,260	0.3%
9	364	882	58.7%
10	1	1	0%
Total	5,485	6,051	9.4%

(color and count) and challenge events in order, each flip as a (deck, color) pair *in order*, and a discard event naming the discarding seat but, by rule, not the discarded color.

Two features distinguish this object from the dynamics state. First, bids enter it: the dynamics state forgets a bid chain the moment the round resolves, but the history keeps every bid ever made. Second, flip *order* enters it: Lemma 5.3 proved order irrelevant to the state, but the sequence of deck choices is observed, and in G an observation is part of the decision point whether or not it moves a counter. Order irrelevance collapses the machine but not the information.

The history tree composes round records over the layered DAG of Figure 6, with the loser-dependent starter rule coupling consecutive rounds. Counting it is a finite recurrence over post-discard hand sizes (n_0, n_1) and the starter; terminal histories, decision histories (nodes at which some seat acts: a reorder, a bid-or-challenge point, a flip-target point, a discard), and boundary histories are each counted by the same traversal. The recurrence and its closed subterms (seat-capped flip sequences, chain-prefix counts by parity) are implemented in `real_game_size.py`; Table 6 reports the exact results.

At the full hand size the exact values deserve to be written out once, because “ 10^{34} ” undersells the fact that they are counted, not estimated:

public decision histories of G : 53,366,481,858,618,743,315,771,794,534,337,111
perfect-recall infosets (both seats) : 2,448,351,139,319,171,077,272,326,424,233,304

The opening round alone shows the state-versus-history gap. The round graph of Section 5 has 119,036 lattice-collapsed resolutions, but 1,888,256 order-sensitive contest records: a factor

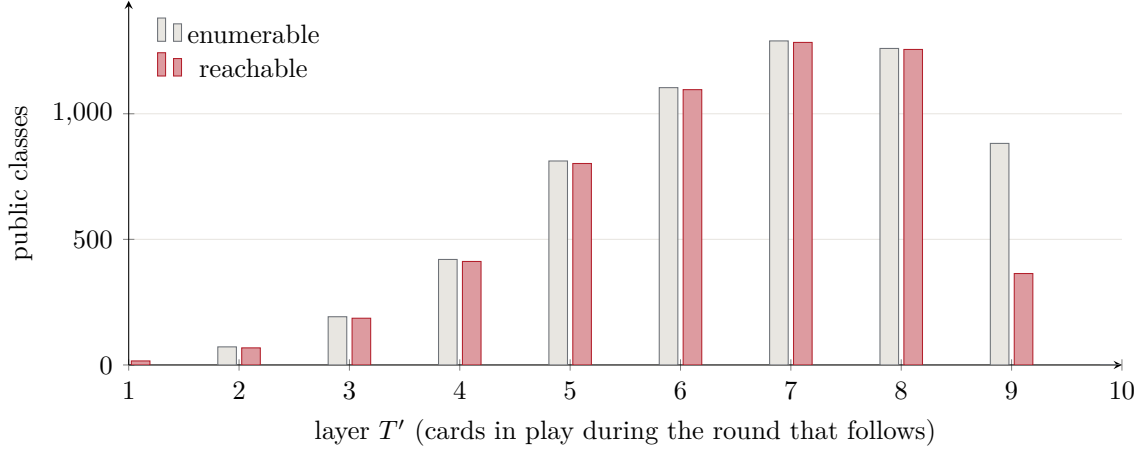


Figure 5: Enumerable versus reachable public classes by layer. The two series are nearly identical everywhere except $T' = 9$, the layer reached after exactly one round, where Corollary 8.3 eliminates 58.7% of the classes the definition admits.

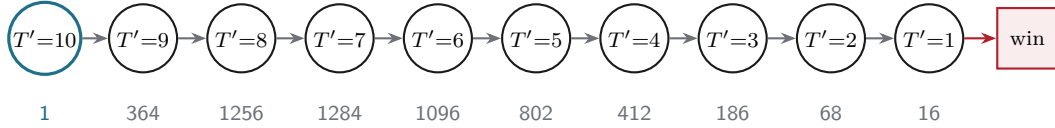


Figure 6: The layered class DAG of the full game. Play enters at the unique root and descends exactly one layer per round, since every round removes exactly one card; the game therefore terminates after between 5 and 9 rounds (Theorem 4.8). Figures beneath each node are the reachable class counts of Table 5.

of 15.9 from flip order and deck choice alone, before bid chains and cross-round composition multiply it to 10^{34} .

9.2 Infosets, exactly

Definition 9.2 (Infoset of G). An infoset of seat i is a distinct pair (public history at one of i 's action points, i 's private trajectory), where the trajectory is i 's initial color multiset, every ordering string i has chosen at a reorder, and every discard color i has chosen. This is exactly what perfect recall retains: i 's own actions and observations, and nothing of the opponent's hidden ones.

Counting pairs is harder than counting histories, because a pair must be *realizable*: consistent with i 's own reveals, and achievable by at least one opponent trajectory. The first constraint is a binomial prefix-consistency count (an ordering whose revealed prefix is fixed leaves $\binom{n-f}{r-p_r}$ free completions). The second reduces to the support theorem of the next section.

Theorem 9.3 (Exact infoset counts). *For heads-up Red/Black at hand size H , the number of perfect-recall infosets per seat is exactly as reported in Table 6; at $H = 5$ the opening seat has 1,224,175,569,659,585,538,636,163,212,116,668 infosets and the responding seat 32 fewer.*

Proof sketch. Walk rounds with state (own exact hand, opponent hand size, the public certainty bounds on the opponent, starter). Own-trajectory multiplicity per round is the product of

Table 6: The real game G , exactly. Public histories are counted by the recurrence of Section 9; per-seat infosets by Theorem 9.3. All values are exact integers computed by `real_game_size.py`; scientific notation is shown for width. The public classes row repeats Corollary 6.7’s closed form $2K(H)^2 + 1$ for comparison.

	$H = 1$	$H = 2$	$H = 3$	$H = 4$	$H = 5$
Terminal public histories	32	1.217×10^6	1.522×10^{13}	4.508×10^{22}	3.023×10^{34}
Public decision histories	55	2.148×10^6	2.687×10^{13}	7.958×10^{22}	5.337×10^{34}
Boundary histories	32	1.254×10^6	1.569×10^{13}	4.647×10^{22}	3.116×10^{34}
Infosets, per seat (starter)	42	945,053	5.932×10^{12}	6.724×10^{21}	1.224×10^{33}
Infosets, both seats	82	1.890×10^6	1.186×10^{13}	1.345×10^{22}	2.448×10^{33}
Public classes ($2K^2+1$)	19	163	723	2,313	6,051

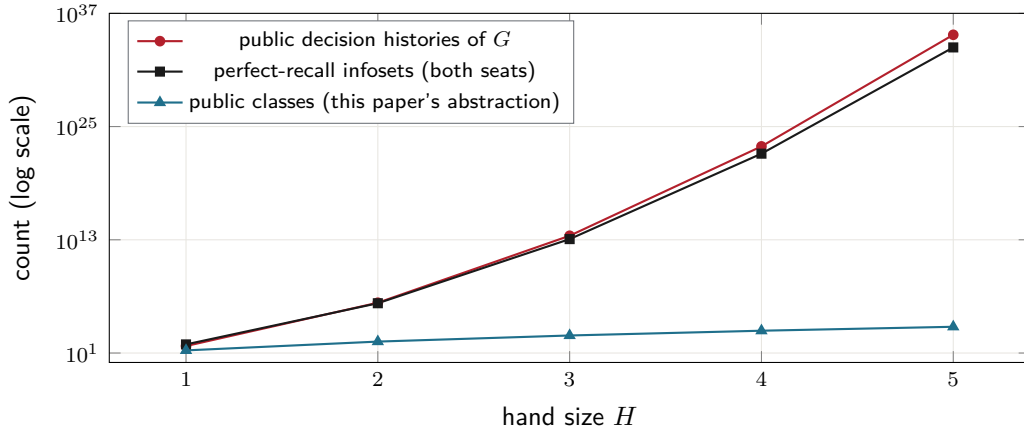


Figure 7: Thirty orders of magnitude between the game and its class abstraction. The history tree and the infoset space of G grow super-exponentially in hand size; the public class space grows like the polynomial $2\left(\binom{H+3}{3} - 1\right)^2 + 1$. Everything the class does not carry must be reconstructed as belief (Section 10).

the ordering-consistency binomial and the discard branching; opponent-side realizability of a candidate reveal (p_r, p_b) reduces, by Theorem 10.1 (support completeness), to non-emptiness of the folded bound interval: $\max(r_{lb}, p_r) + \max(b_{lb}, p_b) \leq n_{opp}$. Because the bounds are *exactly* the feasible set, tight as well as sound, this test admits every realizable pair and no unrealizable one, so the recurrence counts infosets exactly rather than bounding them. The recurrence is validated against explicit joint enumeration of the full game (both hands, both orderings, every deck choice) at $H = 1$ (42 and 40, exact) and $H = 2$ (945,053 and 945,049, exact). $\square$

Remark 9.4 (The starter’s edge, in counts). The two seats’ infoset counts differ by exactly 2 at $H = 1$, 4 at $H = 2$, and 32 at $H = 5$: the opener acts at one node the responder never sees (the opening bid), and the difference compounds only through the thin set of lines where that node is the last asymmetric thing that happens. The game value asymmetry of Paper II has a combinatorial shadow this small.

9.3 What the numbers license

Three ratios summarize the section. The class abstraction compresses boundary histories by about 10^{31} (3.1×10^{34} boundary histories to 5,485 reachable classes, through the intermediate 27,990 concrete boundary states of Table 8). The info set space of G exceeds the class space by about 10^{29} . And the info set space exceeds anything enumerable by any machine by more than ten orders of magnitude. There will be no solve of G by tabulation, at this hand size or anywhere near it: any method that engages G itself must represent histories through a summary that is small and strategically sufficient. Theorem 10.1 will make the class the right summary of public *knowledge*; whether anything tractable sufficiently summarizes public *belief* is the subject of Section 10.

10 What the Class Forgets, and the Belief Family

The public class of Definition 6.1 is a summary, and summaries discard things. This section says exactly what it discards. The omissions are the reason Paper II’s decomposition needs care, and the raw material of Paper III.

10.1 Knowledge versus belief

There is a sharp answer to what the class keeps.

Theorem 10.1 (Support completeness). *Fix a reachable class (ℓ, s_0, s_1) . The hand compositions consistent with some history reaching it are exactly*

$$\{(r_i, b_i) : r_i \geq r_{\text{lb}}^i, b_i \geq b_{\text{lb}}^i, r_i + b_i = n_i, i = 0, 1\}.$$

That is, the certainty bounds are simultaneously sound (nothing outside the box is possible) and tight (everything inside it occurs).

Soundness is Lemma 6.2. Tightness is verified exhaustively: across all 5,485 reachable classes at $H = 5$, the hand compositions actually realized by the closure of Section 8.3 fill the box exactly, with no class showing a gap.

So the class is a complete record of what is publicly *known*, and no finer strategy-free record exists. The exact public posterior depends on the players’ strategies (if a player always discards a black when holding one, watching a discard is enormously informative; against an unknown policy the same observation says almost nothing), so it is a function of the history *and* an opponent model, and cannot be tabulated as part of a description of the game. The certainty bounds are precisely the facts entailed by the rules and the public record regardless of how anyone plays, and Theorem 10.1 says they are *all* of those facts. What the class does not record is what is publicly *believed*; the rest of this section shows that gap is large.

10.2 Two paths to the same class

Consider a seat that ends up holding four cards, certainly including at least one red, public statistic $(4, 1, 0)$. Two histories reach it:

Path A (decay). The seat held five cards; *two reds were flipped off its deck* in round 1. It lost, and discarded. The bound rose to 2, then decayed by one: $r_{\text{lb}} = 1$, $n = 4$.

Path B (fresh sighting). The seat held five cards and nothing was revealed from it in round 1. It lost and discarded, leaving four cards and no bounds. In round 2 *one red* was flipped off its deck: $r_{1b} = 1$, $n = 4$.

Identical classes. But an observer who watched Path A saw two reds go into that hand and only one card leave it, so it very likely still holds two. An observer of Path B saw one red and nothing else. Table 7 quantifies the gap under a neutral reference policy (uniformly random ordering and uniformly random discard), chosen only to put numbers on it.

Table 7: Posterior over the reds a seat holds, given the *same* public statistic $(n, r_{1b}, b_{1b}) = (4, 1, 0)$. Both paths certify “at least one red” and neither certifies more, the bound being tight for each, yet the distributions differ substantially.

Reds held	Path A (two seen, then discard)	Path B (discard, then one seen)
1	0.0531	0.1248
2	0.3184	0.3902
3	0.4521	0.3745
4	0.1765	0.1104
Expected reds	2.752	2.471
Pr[at least 2 reds]	0.947	0.875

The mechanism is the max-fold of Lemma 6.2. Folding remembers the height of the bound but forgets whether that height is fresh evidence or the residue of a taller bound worn down by discards. A bound of 1 that used to be 3 is a much stronger signal than a bound of 1 that has always been 1, and the class cannot tell them apart.

10.3 The bid history

The second omission is blunter: a round’s bid history is cleared when the round ends (Section 2.3) and appears nowhere in the class. Yet bids are chosen by players who can see their own cards, so a bid history is evidence about the bidder’s hand, and a *sequence* of bid histories is evidence about how that player bids: how readily they bluff, how they price a raise.

Neither kind of evidence survives into the public class. Note the asymmetry with the flip record: flips are physical reveals and enter the class as bounds; bids are cheap talk and enter not at all. A player who opened 5R last round and was caught has told the table something real, and the class forgets it completely.

10.4 The reachable belief family

The omissions above say what beliefs the class loses. The positive counterpart is that the lost content has a small, game-determined shape: under *any* strategy profile, the posterior over both seats’ private situations jointly lives in a family whose form is a property of the game, not of the profile.

Write seat i ’s private *type* at a boundary as $t_i = (r_i, x_i)$: the reds currently held and the reds among i ’s own discards. (This is the private statistic the companion papers solve over; what matters here is only its structure.)

Lemma 10.2 (Deal invariance). *Let $a_i = r_i + x_i$. Then a_i is constant along every play: it equals the number of reds dealt to seat i . Flips do not change it (revealed cards stay in the hand), a red discard moves one unit from r_i to x_i , and a black discard changes neither.*

Theorem 10.3 (Rank-one factorization of reachable beliefs). *Fix any strategy profile and any public history h ending at a boundary of class κ . The joint posterior over type pairs factorizes as*

$$\mu_h(t_0, t_1) \propto \lambda_\kappa(t_0, t_1) \cdot g_0(t_0) \cdot g_1(t_1),$$

where λ_κ is the deal-and-bounds prior of class κ and g_i is a per-seat nonnegative weight (a likelihood tilt) depending only on seat i 's strategy and h .

Proof. All correlation between the seats' private situations originates in the deal. By Lemma 10.2 the deal enters the type pair only through (a_0, a_1) , which is a deterministic function of the types themselves; the deal coupling is therefore absorbed into λ_κ and rides along every transition unchanged. Conditional on types, each seat's observable actions depend only on its own type and h , so the likelihood of h is a product of per-seat terms. Bayes' rule then yields the displayed form. $\square$

Corollary 10.4 (Model-free conditionals). *The conditional law of seat i 's type given the opponent's type, $\mu_h(t_i \mid t_{1-i})$, does not depend on g_{1-i} : any positive weighting of opponent types produces the same conditionals. An observer can be entirely wrong about how the opponent plays and still hold exactly correct own-side conditionals.*

Theorem 10.3 was found empirically before it was proved: over 17,140 boundaries of 3,000 self-play games under the solved profile of the companion papers, the measured rank-one residual of the exact posterior is 3.3×10^{-16} , floating-point zero, with no exceptions. The theorem is load-bearing there twice over: it makes exact belief tracking cheap (two per-seat vectors instead of a joint matrix), and Corollary 10.4 is what lets a re-solving agent keep exact own-side conditionals even against an unmodeled opponent (Paper III's filter-repair construction).

Together with Section 9, this completes the ladder of Section 3. The public class is the strategy-independent *knowledge*; the reachable *belief* state is the class plus two per-seat tilt vectors (g_0, g_1) , a finite-dimensional family against the 10^{34} raw histories that induce it. Strategy dependence enters only through where in the family a profile lands, never through the family's shape.

Remark 10.5 (The deal coupling in numbers). Because the deck is finite and balanced, a seat's own cards are evidence about the opponent's: seeing k reds among your own five shifts the chance that an unseen card is red from $1/2$ to $(26 - k)/47$, a swing from 0.447 to 0.553. In the one-card endgame (both seats down to one card, layer $T' = 2$) this coupling is the entire strategic content: the opener's only real decision is whether to bet that the two remaining cards match or differ, using its own five dealt cards to choose. That endgame's exact value is $\frac{40651}{78302} \approx 0.5192$ to the opener, above $1/2$ purely because of sampling without replacement, and above the $26/51 \approx 0.5098$ that an opener ignoring its own cards would get. Under an idealised infinite deck the same game is exactly even.

10.5 Measured: how much the class forgets

The gap of Table 7 was a two-path worked example. Under the solved profile of the companion papers it has been measured wholesale: the reach-weighted total-variation distance between

the class prior λ_κ and the true public posterior μ_h is **0.842** over self-play boundaries, rising monotonically with depth from 0.66 (first boundary) to 0.95 (endgame layers). Replacing the prior with the best possible *class-keyed* statistic closes only about 17 points of that gap, and small public-statistic enrichments (e.g. freshness bits distinguishing Path A from Path B above) add only 2–3 more: most of the belief lives in within-class variation that no small public statistic reaches. These are profile-dependent numbers, measured under one representative profile, the solved one, but they put a scale on the qualitative claim of this section: the class prior at depth is nearly orthogonal to the belief a competent observer actually holds.

10.6 The abstraction G'

The companion papers divide along the ladder’s last rung, and the split object deserves a formal statement.

Definition 10.6 (The abstraction G'). G' is the game identical to G within each round, except that at every round boundary of class κ the joint law over type pairs is reset to the class prior λ_κ of Theorem 10.3; equivalently, the tilts g_0 and g_1 are replaced by 1. Its decision points are pairs (public class, own type) rather than histories, so its state notion is rung (iii) of Section 3.

Paper II solves G' exactly, decomposing at boundaries over the 5,721 subgame-bearing classes, and then measures what the reset costs in play: the abstraction’s equilibrium is exploitable in G in ways its own convergence meters cannot see. Paper III keeps the tilts, tracking (g_0, g_1) with an exact public filter, re-solving each boundary under the true posterior rather than the prior, and pursuing a certified upper bound on real-game exploitability, with Corollary 10.4 supplying own-side conditionals that survive an arbitrarily wrong opponent model. The distance between the two programs is the measured gap of Section 10.5. A standalone companion paper on solver coupling (in preparation) records where the two programs met: the coupled store `solve-6-cfrpolicy`, whose quantal-response values are held fixed while per-class CFR+ policies are re-solved against them. That store is the program’s endpoint, the first that Paper III’s belief-carrying probe cannot beat: its real-game exploitability lower bound (the quantity ϵ_G of Papers II and III) reads -0.025 at $n = 10,000$ games per arm (artifact `pool_solve-6-cfrpolicy_vs_solve2_full.json`).

Remark 10.7 (Program endpoint, 2026-08). The bound of record, measured by the probe of Paper III (`probe_resolver.py`, guards on, matched seeds): the coupled store `solve-6-cfrpolicy` reads $\epsilon_G \geq -0.025$ $[-0.044, -0.005]$ at $n = 10,000$ games per arm, against the incumbent store’s $+0.034$ $[+0.015, +0.054]$ at the same n and seeds. The probe loses to it, so the bound is negative and therefore vacuous as a bound; what the measurement supports is the comparison, which resolves at $p = 3.0 \times 10^{-5}$. The artifact is `pool_solve-6-cfrpolicy_vs_solve2_full.json`. The enumeration in this paper is untouched by any of this; the remark is here because the ladder of Section 3 is the frame the endpoint is stated in (the probe carries rung-(iv) beliefs against a rung-(iii) store).

11 Summary of Counts

Table 8 gathers the counts across the three games. The deal count climbs nine orders of magnitude and the opening round’s contest structure four, while the public class space grows only from 19 to 6,051; the histories and infosets of the true game (Table 6) grow past 10^{33} .

Table 8: Master table. Every entry is produced by `state_counts.py` and, where an engine counterpart exists, verified against the shipped implementation by `verify_against_engine.py`.

Quantity	$H = 1$	$H = 2$	$H = 5$
Card-level deals	2,652	1,624,350	3,986,646,103,440
Color-level deal outcomes	4	9	36
Rounds (min–max)	1–1	2–3	5–9
Opening-round bid states	7	31	2,047
Opening-round procurement	14	190	65,790
Opening-round resolutions	28	348	119,036
Per-seat public statistic $K(H)$	3	9	55
Public classes $2K^2 + 1$	19	163	6,051
of which terminal	18	54	330
of which reachable	17	149	5,485
Concrete boundary states	28	345	27,990
Public decision histories of G	55	2,147,639	5.337×10^{34}
Perfect-recall infosets of G (both seats)	82	1,890,102	2.448×10^{33}

A Reproducing the numbers

Three scripts accompany this paper.

state_counts.py re-derives every class-space count from the rules, independently of the game engine, and prints the corresponding tables. It includes the forward-closure search of Section 8.2.

real_game_size.py computes every count in Section 9 (histories, decision histories, boundary histories, and the exact per-seat infoset counts) as exact big integers, and runs the brute-force joint-enumeration validation of Theorem 9.3 at $H = 1, 2$. Pure standard library.

verify_against_engine.py cross-checks the round-graph counts, the class-space counts and the reachable set against the shipped implementation, so any disagreement is a genuine bug in one of the two rather than a difference of convention.

Run them with

```
cd research && python3 state_counts.py
cd research && python3 real_game_size.py
cd research && ../simulator/.venv/bin/python verify_against_engine.py
```

B Glossary

Boundary The moment a contest resolves: loser known, discard not yet made. The natural cut point for describing the public state.

Chain The strictly increasing sequence of counts bid in a round.

Class The public state at a boundary (Definition 6.1).

Layer T' Total cards in play after the pending discard; strictly decreasing, hence a valid induction index.

Procurement The contest phase: the bidder flips cards trying to produce the claim.

Sighting The multiset of colors revealed from one seat's deck in one round.